

ROAD DAMAGE DETECTION AND CRACK SEGMENTATION USING DEEP LEARNING

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PRACTICE SCHOOL-II

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OUTLINE OF THE PRESENTATION

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- Objectives
- Methodology
- Detection Experiments
- SAHI Evaluation
- Segmentation Experiments
- Recent Literature Comparison
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PROJECT OVERVIEW

About the Project

Project Title: Road Damage Detection and Crack Segmentation using Deep Learning

Project Type: Bachelor's Thesis / Practice School-II

Research Area: Computer Vision and Deep Learning

Datasets: RDD2022 and Crack500

Research Scope

- Object Detection
- SAHI Evaluation
- Crack Segmentation

Faculty Mentor Details

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PROJECT TIMELINE

Start date: 1st June, 2026

Project Milestones

Phase 1: Literature review and problem understanding completed.

Phase 2: Dataset preparation and exploratory data analysis completed.

Phase 3: YOLOv8 and YOLO11 object detection experiments completed.

Phase 4: SAHI evaluation and YOLO11m ablation study completed.

Phase 5: Faster R-CNN implementation and exploratory SAHI evaluation completed.

Phase 6: Crack segmentation using U-Net and DeepLabV3+ completed.

Phase 7: Comparison with recent literature and result analysis completed.

Phase 8: Documentation, thesis writing and presentation preparation in progress.

End date: 26th July, 2026

PROBLEM IDENTIFICATION

Statement: Road surface defects such as cracks and potholes reduce driving safety and increase road maintenance costs. Manual inspection is slow, expensive, and often inconsistent. An automated computer vision-based system can provide faster and more reliable road damage assessment.

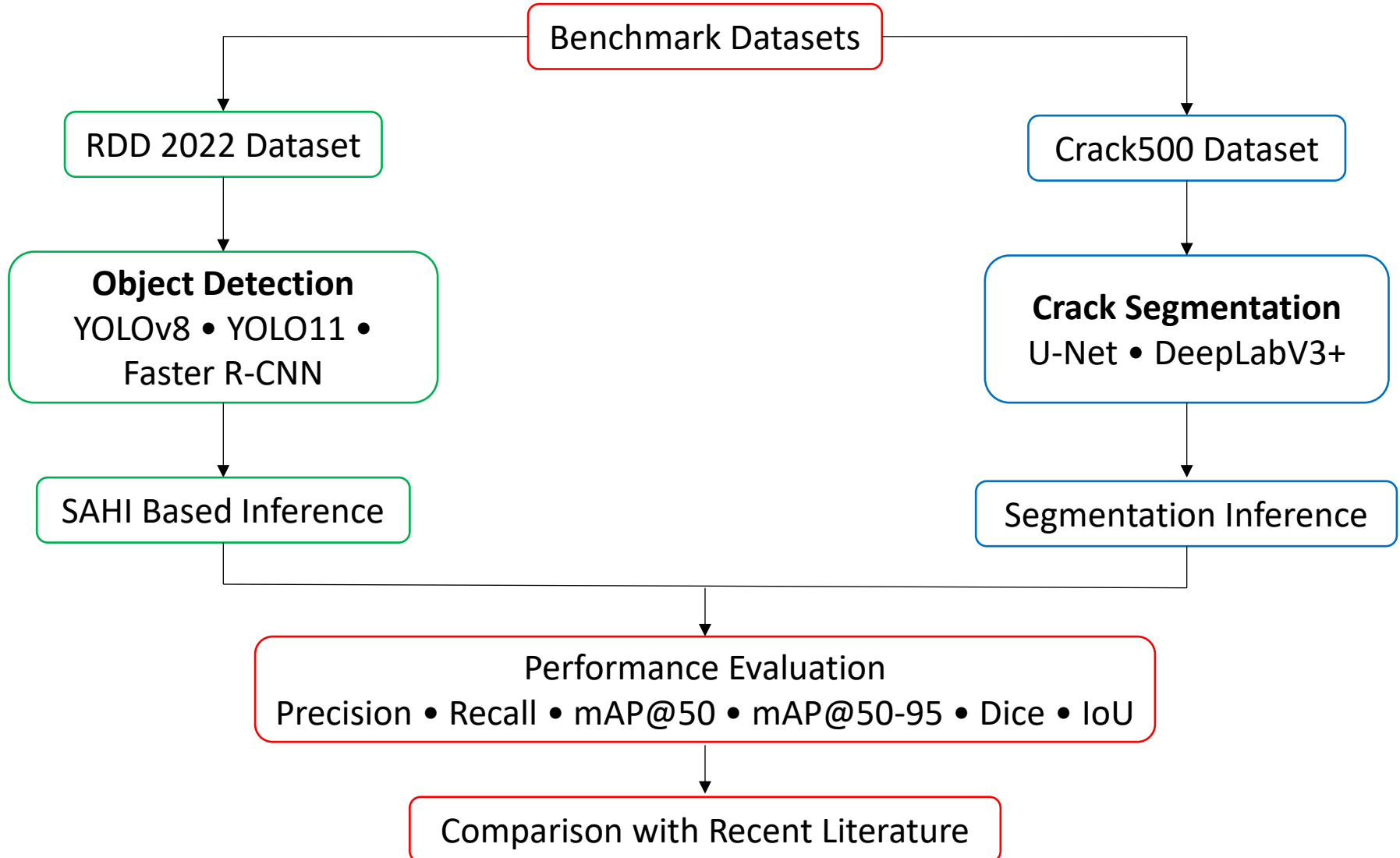
Objectives:

- Evaluate multiple deep learning models for road damage detection.
- Study the impact of SAHI on small-object detection.
- Compare U-Net and DeepLabV3+ for crack segmentation.
- Benchmark the proposed models against recent literature.

Methodology:

- Road Damage Detection using **YOLOv8, YOLO11 and Faster R-CNN**
- SAHI-based sliced inference evaluation
- Crack Segmentation using **U-Net and DeepLabV3+**
- Performance evaluation using **Precision, Recall, mAP, Dice Score and IoU**

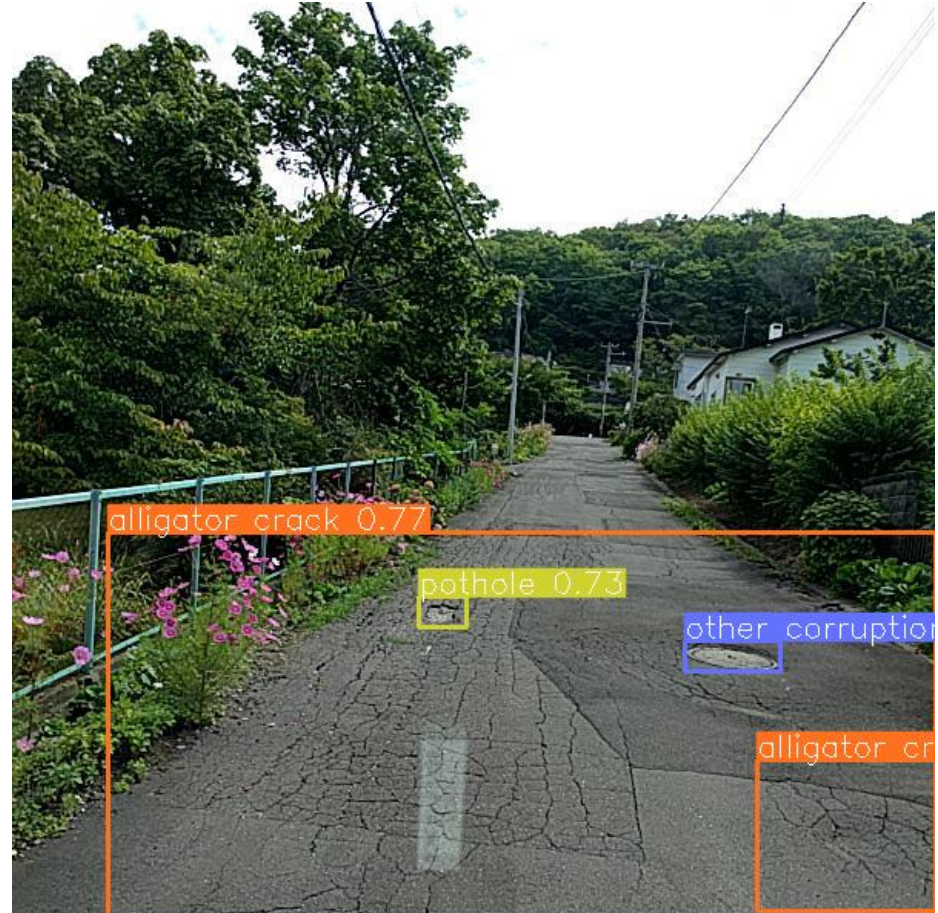
METHODOLOGY



DETECTION EXPERIMENTS

Model	mAP@50	mAP@50-95
YOLOv8n	0.539	0.285
YOLOv8s	0.592	0.317
YOLOv8m	0.633	0.344
YOLOv11n	0.544	0.287
YOLOv11s	0.575	0.309
YOLOv11m	0.587	0.314
Faster R-CNN	0.536	0.253

Baseline performance comparison of all evaluated detection models.



Representative YOLOv8m predictions on the RDD2022 dataset.

SAHI EVALUATION

Model	Baseline Detection	SAHI Detection	Detection Increase
YOLOv8n	3773	4564	20.96%
YOLOv8s	4237	5475	29.22%
YOLOv8m	4551	5321	16.92%
YOLOv11n	3612	4496	24.47%
YOLOv11s	4016	4826	20.17%
YOLOv11m	3986	4722	18.46%

- SAHI increased detections for every YOLO model.
- YOLOv8s showed the largest improvement (**29.22%**).
- The increase in detections was accompanied by additional false positives.



Baseline YOLO vs. YOLO + SAHI Predictions

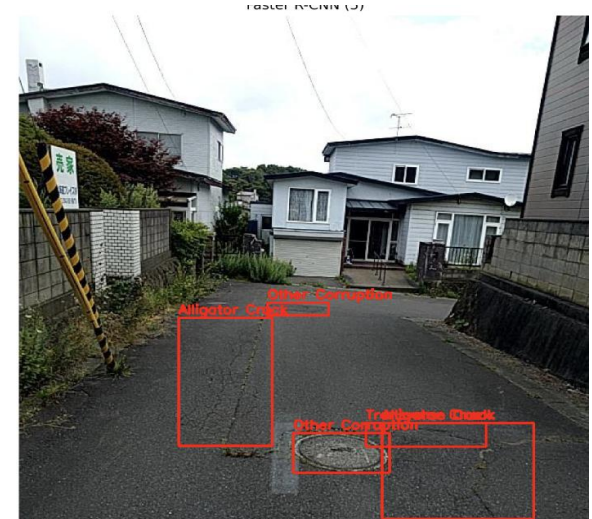


EXPLORATORY STUDY: FASTER R-CNN + SAHI

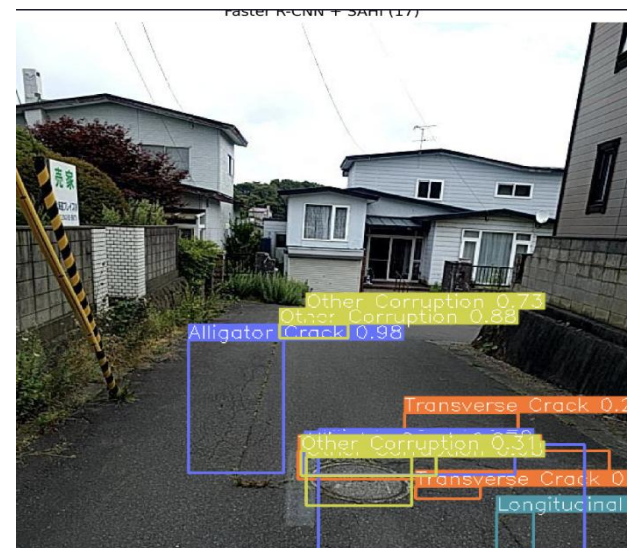
Model	Baseline Detection	SAHI Detection	Detection Increase
Faster R-CNN	6899	11256	63.15%

Metrics	Value
Precision	0.203
Recall	0.271
mAP@50	0.203
mAP@50-95	0.086

- SAHI increased detections by **63.15%**.
- Many of the additional detections were false positives and duplicate boxes.
- Therefore, Faster R-CNN + SAHI was treated as an **exploratory study** rather than the final approach.



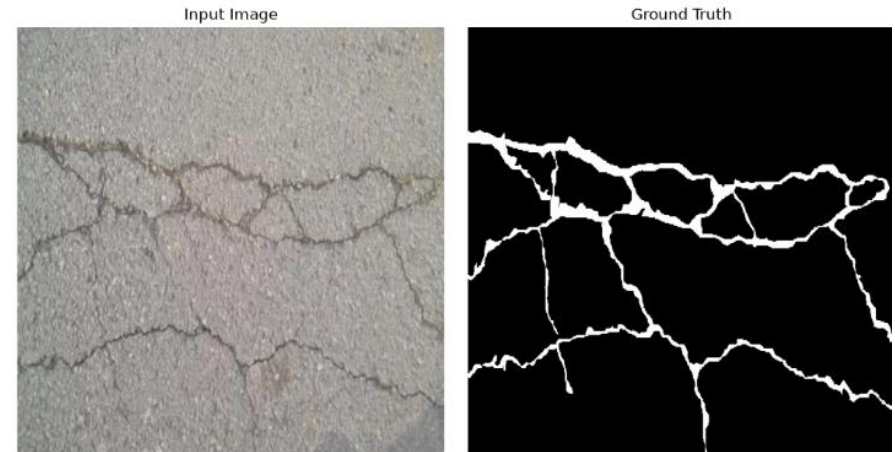
Example showing additional detections after applying SAHI.



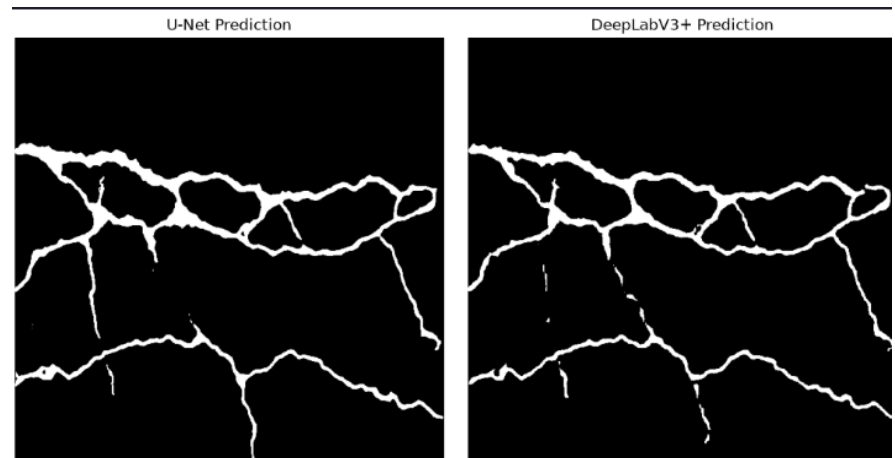
SEGMENTATION EXPERIMENTS

Model	Dice Score	IoU Score
U-Net	0.7533	0.6118
DeepLabV3+	0.7389	0.5931

- U-Net achieved the highest Dice Score (**0.7533**) and IoU (**0.6118**).
- Both models successfully segmented the primary crack regions..
- U-Net preserved thin crack structures better and was selected as the final segmentation model.



Segmentation results on a Crack500 sample image.



DETECTION LITERATURE COMPARISON

Method	Year	Dataset	mAP@50	mAP@50-95
RDD-YOLO	2024	RDD2022	62.5	36.4
Improved YOLOv8	2024	RDD2022	65.7	—
YOLOv8-PD	2024	RDD2022	70.6	39.5
YOLO-RD*	2025	Japan Subset	55.62	25.75
OBC-YOLOv8†	2025	China Subset	86.0	—
SEA-YOLOv8	2025	RDD2022	63.2	—
YOLOv8m	2026	RDD2022	63.3	34.4
YOLOv8m + SAHI	2026	RDD2022	48.4	24.7

YOLO-RD was evaluated on the Japan subset of RDD2022. OBC-YOLOv8 was evaluated on the China subset. Some papers report only mAP@50, so mAP@50-95 is unavailable.

Observation: YOLOv8m achieved competitive performance on the RDD2022 benchmark despite using a standard training pipeline without architectural modifications.

SEGMENTATION LITERATURE COMPARISON

Method	Year	Dataset	Dice (%)	IoU (%)
Pyramid Attention Network	2020	Crack500	76.81	62.35
Dual Flow Fusion Model*	2023	Crack500	78.10	66.00
Distribution-aware Noisy-label Learning	2024	Crack500	74.74	60.56
EGA-UNet†	2025	Crack500	77.80	70.00
U-Net	2026	Crack500	75.33	61.18
DeepLabV3+	2026	Crack500	73.89	59.31

The Dual Flow Fusion paper reports F1-score, which is equivalent to the Dice coefficient for binary segmentation. EGA-UNet reports foreground IoU, which is directly comparable to the IoU used in this work.

Observation: U-Net achieved competitive performance on the Crack500 dataset and outperformed DeepLabV3+ in both Dice Score and IoU.

CONCLUSIONS

Key Findings

- Evaluated seven object detection models on the RDD2022 dataset.
- YOLOv8m achieved the best overall detection performance with 63.3% mAP@50 and 34.4% mAP@50-95.
- SAHI increased the number of detected road damages for all YOLO models, with a maximum improvement of 29.22%.
- The Faster R-CNN + SAHI study showed a 63.15% increase in detections but also introduced many false positives.
- U-Net achieved the best crack segmentation performance with a Dice Score of 75.33% and an IoU of 61.18%.
- Comparison with recent literature showed that the proposed models achieved competitive performance on benchmark datasets.

LEARNING OUTCOMES

Technical Skills

- Gained hands-on experience with object detection and semantic segmentation.
- Compared one-stage (YOLO) and two-stage (Faster R-CNN) detection models.
- Evaluated the impact of SAHI on small-object detection.
- Performed model evaluation using Precision, Recall, mAP, Dice Score, and IoU.

Research Skills

- Conducted a comprehensive literature review.
- Designed and compared multiple deep learning experiments.
- Analyzed experimental results and benchmarked them against published research.
- Improved technical writing and presentation of research findings.

REFERENCES

1. Arya, D., *et al.*, **RDD2022: A Multi-national Image Dataset for Automatic Road Damage Detection**, *Scientific Data*, 2022.
2. Akyon, F. C., Altinuc, S. O., and Temizel, A., **Slicing Aided Hyper Inference and Fine-Tuning for Small Object Detection (SAHI)**, *ICIP*, 2022.
3. Wang, C.-Y., Bochkovskiy, A., and Liao, H.-Y. M., **YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors**, 2022.
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6. Chen, L.-C., *et al.*, **Encoder–Decoder with Atrous Separable Convolution for Semantic Image Segmentation (DeepLabV3+)**, *ECCV*, 2018.
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8. Zeng, J., and Zhong, H., **YOLOv8-PD: An Improved Road Damage Detection Algorithm Based on YOLOv8n**, *Scientific Reports*, 2024.
9. Wang, W., *et al.*, **YOLO-RD: A Road Damage Detection Method for Effective Pavement Maintenance**, *Sensors*, 2025.
10. Zhang, S., *et al.*, **OBC-YOLOv8: An Improved Road Damage Detection Model Based on YOLOv8**, *PeerJ Computer Science*, 2025.

Additional references used for the literature review are provided in the accompanying Summary Findings document.

COMPLETE DETECTION PERFORMANCE

Model	Precision	Recall	mAP@50	mAP@50-95
YOLOv8n	0.579	0.516	0.539	0.285
YOLOv8s	0.626	0.553	0.592	0.317
YOLOv8m	0.677	0.590	0.633	0.344
YOLO11n	0.593	0.514	0.544	0.287
YOLO11s	0.613	0.540	0.575	0.309
YOLO11m	0.631	0.546	0.587	0.314
Faster R-CNN	N/A	0.427	0.536	0.253

Precision is marked N/A because it was not computed during the Faster R-CNN baseline evaluation; the reported Recall and mAP values are from the COCO evaluator.

Observation: YOLOv8m achieved the highest overall detection performance across all evaluated models, obtaining the best Precision (0.677), Recall (0.590), mAP@50 (0.633), and mAP@50-95 (0.344).

COMPLETE SAHI PERFORMANCE

Model	Precision	Recall	mAP@50	mAP@50-95
YOLOv8n	0.380	0.285	0.380	0.186
YOLOv8s	0.441	0.311	0.441	0.218
YOLOv8m	0.484	0.347	0.484	0.247
YOLO11n	0.380	0.277	0.380	0.184
YOLO11s	0.421	0.301	0.421	0.207
YOLO11m	0.412	0.301	0.412	0.203

The top table presents the complete SAHI evaluation across all YOLO models, while the bottom table shows the YOLO11m ablation study used to determine the optimal SAHI configuration

Slice Size	Confidence	Precision	Recall	mAP@50	mAP@50-95
512×512	0.25	0.412	0.301	0.412	0.203
512×512	0.50	0.307	0.218	0.307	0.171
640×640	0.25	0.402	0.299	0.402	0.208

- SAHI improved small-object detection across all YOLO models.
- Additional detections came with increased false positives.
- YOLOv8m remained the best-performing detector after applying SAHI.
- Default configuration (512×512, confidence = 0.25) provided the best trade-off and was used for all remaining experiments.